

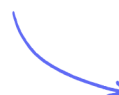
Non-Calculator Questions

Converting Fractions, Decimals & Percentages

Converting Fractions, Decimals & Percentages / Recurring Decimals / Ordering Fractions, Decimals & Percentages

Easy (5 questions)	/11
Medium (5 questions)	/15
Hard (5 questions)	/11
Total Marks	/37

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Easy Questions

- 1 Here is a list of numbers.

Put a ring around the number with the largest value.

$$0.3030 \quad \frac{1}{3} \quad 0.0330 \quad \frac{3}{10} \quad 33\%$$

(1 mark)

- 2 Mr Mason asks 240 Year 11 students what they want to do next year.

15% of the students want to go to college.

$\frac{3}{4}$ of the students want to stay at school.

The rest of the students do not know.

Work out the number of students who do not know.

(4 marks)

- 3 Karen got 32 out of 80 in a maths test. She got 38% in an English test.

Karen wants to know if she got a higher percentage in maths or in English.

Did Karen get a higher percentage in maths or in English?

(2 marks)

4 Prove algebraically that the recurring decimal $0.2\dot{5}$ has the value $\frac{23}{90}$

(2 marks)

5 Show that the recurring decimal $0.1\dot{7} = \frac{8}{45}$

(2 marks)

Medium Questions

- 1 Mr Brown and his 2 children are going to London by train.

An adult ticket costs £24

A child ticket costs £12

Mr Brown has a Family Railcard.

Family Railcard gives

$\frac{1}{3}$ off adult tickets

60% off child tickets

Work out the total cost of the tickets when Mr Brown uses his Family Railcard.

(4 marks)

- 2 Write these numbers in order of size.
Start with the smallest number.

0.24 $\ddot{6}$

0.24 $\dot{6}$

0. $\dot{2}$ 4 $\ddot{6}$

0.246

(2 marks)

3 Show that the recurring decimal $0.\dot{3}9\dot{6} = \frac{44}{111}$

(2 marks)

4 Use algebra to show that the recurring decimal $0.\dot{4}1\dot{7} = \frac{139}{333}$

(2 marks)

5 (a) Show that $0.\dot{1}\dot{9}$ can be written as $\frac{19}{99}$.

(3 marks)

(b) Explain how $\frac{19}{99} = 0.\dot{1}\dot{9}$ can be used to find $\frac{19}{990}$ as a decimal and write down its value.

$$\frac{19}{990} = \dots\dots\dots$$

(2 marks)

Hard Questions

- 1 Write the recurring decimal $0.1\dot{7}$ as a fraction in its simplest form. You must show all your working.

(3 marks)

- 2 Write the recurring decimal $0.2\dot{6}$ as a fraction.
You must show all your working.

(2 marks)

- 3 Write the recurring decimal $0.3\dot{2}$ as a fraction.
You must show all your working.

(2 marks)

- 4 Write the recurring decimal $0.6\dot{7}$ as a fraction.
Show all your working and give your answer in its simplest form.

(2 marks)

- 5 Change the recurring decimal $0.1\dot{8}$ to a fraction.
You must show all your working.

(2 marks)